

MAE 4730

10/26/18

Lecture 26

Skate (cont'd)

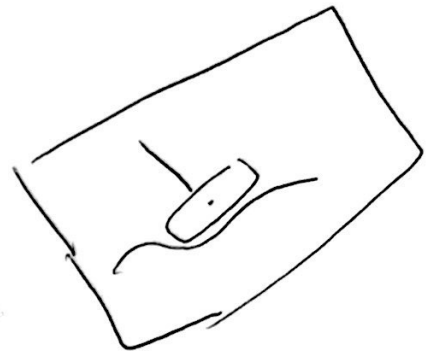
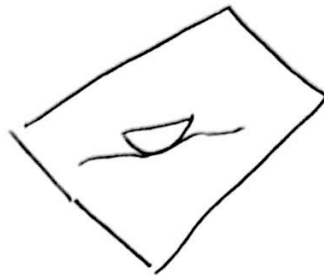
1) General comments

2) DAEs

3) Lagrange

General Comments

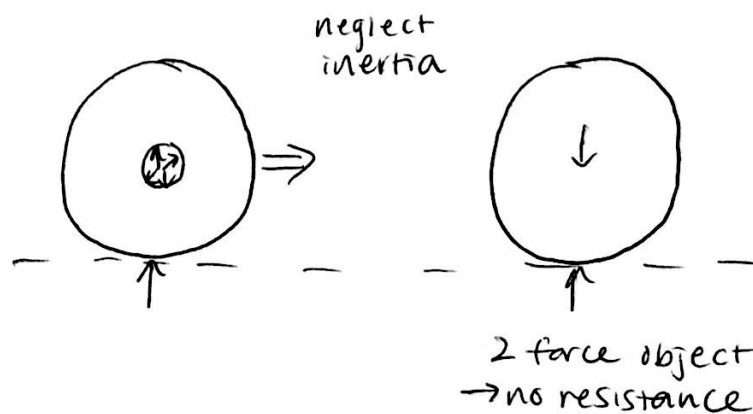
Skate vs. wheel



- Allowed motion the same
- Skate: neglect friction along motion
- Wheel: neglect a) "rolling" friction
b) inertia of wheels

Wheel: Side wheel:

ideal wheel:



- Assume:
 - wheel round
 - negligible friction in bearing
 - point contact w/ ground

Inertia:



$$AMB/c \Rightarrow M/c = \dot{H}/c$$

$$Fr = I\alpha$$

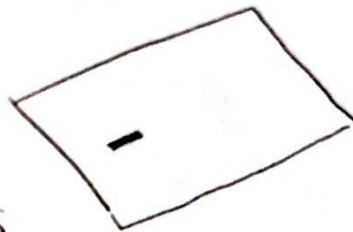
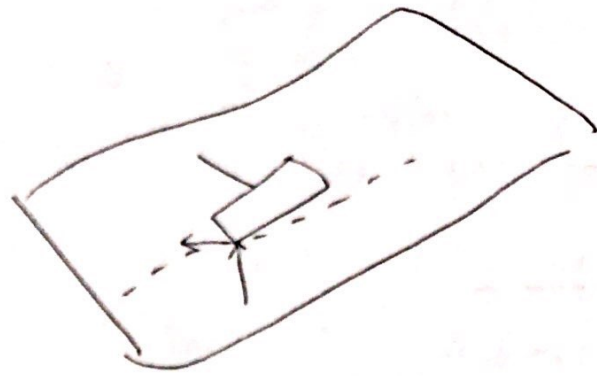
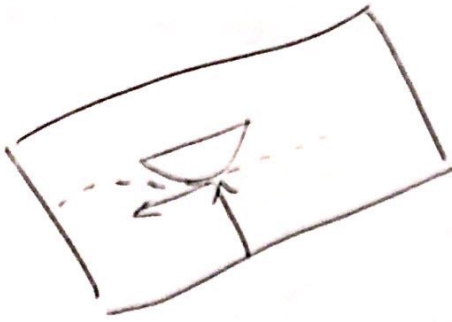
$$\alpha = \frac{a}{r} \leftarrow \ddot{\theta}$$

$$I \approx m_{\text{wheel}} r^2$$

$$a = \frac{F_{\text{ext}}}{m_{\text{car}}}$$

$$\Rightarrow F \approx \frac{F_{\text{car}} m_w}{m_{\text{car}}} \approx 0$$

friction force



Degrees of Freedom:

3 (rigid body in plane)

↳ configuration degrees of freedom

2
↳ velocity degrees of freedom

$3 > 2$: non-holonomic systems

θ, y_c, x_c : "ignorable coordinates"
"symmetry variables"

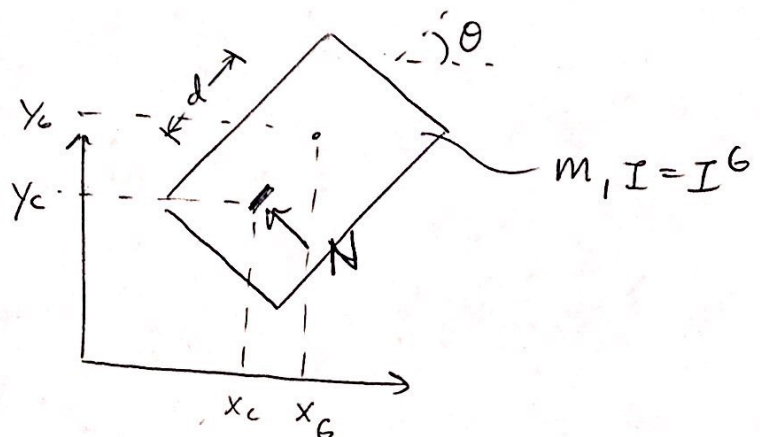
We used $w(t)$, $v_c(t)$

⇒ ODEs ⇒ solutions

DAE = maximal coordinates

Maximal coordinates:

x_G, y_G, θ



LMB: $\sum \vec{F} = m\vec{a}$ (1)

$$-N\sin\theta\hat{i} + N\cos\theta\hat{j} = m(\ddot{x}\hat{i} + \ddot{y}\hat{j})$$

AMB₁₆ $\Rightarrow I\ddot{\theta} = -Nd$

3 eqns for $\ddot{x}_6, \ddot{y}_6, \ddot{\theta}, N$

Constraint eq:

$\vec{v}_c = v_c \hat{\lambda}$ or $\vec{v}_c \cdot \hat{n} = 0$ ✓

Adds new variable

$\vec{v}_c = \vec{v}_G + \vec{v}_{c/G}$

$\vec{v}_G = \dot{x}_6\hat{i} + \dot{y}_6\hat{j}$

$\vec{v}_{c/G} = \vec{\omega} \times \vec{r}_{c/G}$

$\vec{\omega} = \dot{\theta}\hat{k}$

$\vec{r}_{c/G} = -d\hat{\lambda}$

$\hat{\lambda} = \cos\theta\hat{i} + \sin\theta\hat{j}$

$$\hat{n} \cdot \vec{v}_c = -\hat{n} \cdot \vec{\omega} \times (\hat{\lambda}d)$$

$$- \sin\theta\dot{x} + \cos\theta\dot{y} = \dot{\theta}d \quad (4)$$

$$(4) \Rightarrow \frac{d}{dt}(4)$$

$$\Rightarrow (-\cos\theta\dot{\theta}\dot{x} - \sin\theta\ddot{x}) + (-\sin\theta\dot{\theta}\dot{y} + \cos\theta\ddot{y}) = \ddot{\theta}d$$

$$(1) - (4)$$

$$\Rightarrow_{LMB} m\ddot{x} + 0\ddot{y} + 0\ddot{\theta} + -\sin\theta N = 0$$

$$LMB_y \quad 0\ddot{x} + m\ddot{y} + 0\ddot{\theta} + -\cos\theta N = 0$$

$$AMB_{16} \quad 0\ddot{x} + 0\ddot{y} + I\ddot{\theta} + dN = 0$$

$$- \sin\theta\ddot{x} + \cos\theta\ddot{y} + -d\ddot{\theta} + 0N = \cos\theta\dot{\theta}\dot{x} + \sin\theta\dot{\theta}\dot{y}$$

$\Rightarrow$ Can create matrix $\Rightarrow$ MATLAB $\Rightarrow$ MAGIC!

Lagrange Equations:

min coordinates: x, y, θ

Lag eqns: $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = Q_x$

$$\dot{y} = Q_y$$

$$\dot{\theta} = Q_\theta$$

$$\Rightarrow m \ddot{x} = Q_x = -N \sin \theta$$

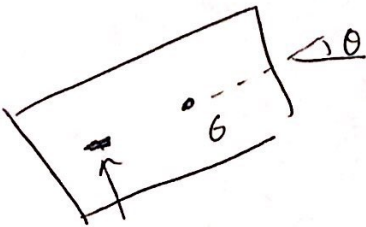
$$m \ddot{y} = Q_y = N \cos \theta$$

$$m \ddot{\theta} = Q_\theta = -dN$$

$$Q_x \Rightarrow Q_x dx = dW$$

$$Q_y \Rightarrow Q_y dy = dW$$

$$Q_\theta \Rightarrow Q_\theta d\theta = dW$$



$$dW = \overset{Q_x}{(-\sin \theta) dx} N + \overset{Q_y}{\cos \theta dy} N + \overset{Q_\theta}{(-d) d\theta} N$$

Supplement Lagrange eqn w/ same const. eqns for DAE

- DAE + Lagrange are identical (set of eqns)